A1_CP414_W2026

Monday, January 19, 2026 9:37 AM

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CP414, Winter 2026 - Assignment 1

Upload **a single PDF file** to MyLS Dropbox. Your assignments in this class will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically.

1. For each of the below parts, draw the state diagram of the **DFA** that recognizes the given language. For full credit, it should have the minimal number of states (there is no need to prove that the number of states is minimal)

0. $L = the\ empty\ language\ \emptyset$ with $\Sigma = \{a, b\}$
1. $L = the\ language\ that\ accepts\ only\ the\ empty\ string\ \varepsilon$ with $\Sigma = \{a, b\}$
2. L = the set of all strings that have a length at least 3 and its third character is a *1* with $\Sigma = \{0, 1\}$

   * : Kleene star , : AND
3. L = $\{w \in \{a, b\}^* \mid w\ contains\ an\ odd\ number\ of\ a's, even\ number\ of\ b's, starts\ with\ an\ a\}$
4. L = The set of all strings that does not contain pattern 0100 with $\Sigma = \{0, 1\}$
5. L = $\{w \in \{0, 1\}^* \mid w\ contains\ exactly\ one\ occurrence\ of\ the\ substring\ 11\}$

   Note: *111* does not belong to L since there are 2 occurrences of the substring *11*
6. L1 = The set of all strings that start with an *a* or have odd length with $\Sigma = \{a, b\}$
7. L2 = The set of all strings that start with a *b* and have even length with $\Sigma = \{a, b\}$
8. L1 ∩ L2
9. The set of all strings over $\Sigma = \{0, 1\}$ such that every occurrence of *1* is followed by at least two *0's*, e.g., **0001000100, 100, 0, 00000000010000000100100** are in this language, but **1011, 1, 101** are not.
10. The set of all strings that does not contain the pattern **0110** over $\Sigma = \{0, 1\}$.
11. The set of all strings except **abb** and **ba** over $\Sigma = \{a, b\}$.

   not contain
2. Consider the following languages **A** and **B**:

$A = \{w \in \{0, 1\}^* \mid every\ 1\ in\ w\ is\ immediately\ followed\ by\ at\ least\ one\ 0\}$

$B = \{w \in \{0, 1\}^* \mid w\ has\ an\ odd\ number\ of\ zeroes\}$

(**a**) Draw the state diagram of the **DFA** of the following language: ***A*** ∪ ***B***. It may help to first construct the **DFA**s for ***A*** and ***B***.

(**b**) Draw the state diagram of the NFA of the following language: **(A)*** ∘ **B**.

3. For each **NFA** below:

(1) start state q_1, accepting state q_2

	a	b	ε
-> q_1	{q_2}	{q1, q2}	Ø
* q_2	{q_1}	Ø	Ø

(2) start state q_1, accepting state q_2

	a	b	ε
-> q_1	Ø	Ø	{q_3}
* q_2	{q_2, q_3}	{q_3}	Ø
q_3	{q_3}	{q_2}	{q_3}

(**a**) Provide an **NFA** state transition diagram.

(**b**) Use the construction given in **Theorem 1.39** to convert the **NFA** to an equivalent **DFA**. Show your work (including **ALL** intermediate steps).

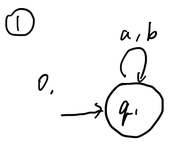
4. In certain programming languages, comments appear between delimiters such as /# and #/. Let C be the language of all valid delimited comment strings. A member of C must begin with /# and end with #/ but have no intervening #/. For simplicity, assume that the alphabet for C is Σ = {a, b, /, #}. Give an NFA that recognizes the language C.

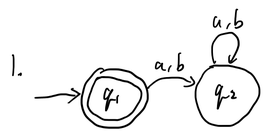
C = { w = {a,b,/,#}* | w start /# & end #/ & no #/ inbetween }

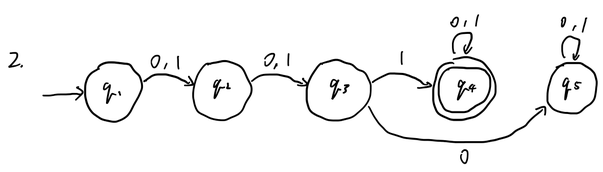
5. Give a nondeterministic finite automaton accepting the set of strings of **0**'s and **1**'s such that there are two **1**'s separated by a number of positions that is a multiple of 3. (Note: 0 is not an allowable multiple of 3, so **11** is not in language while **0100110** is). Try to take advantage of nondeterminism as much as possible.

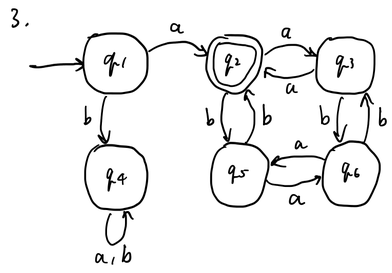
L = { w = {0,1}* | w contain 1 [multiple of 3] 1 }

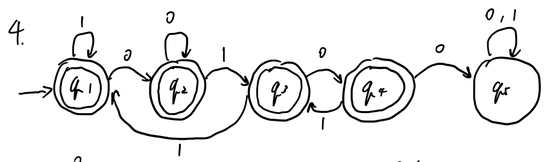
6. Let D = {w | w contains an even number of **a**'s & an odd number of **b**'s & does not contain the substring **ab**}. Give a **DFA** with five states that recognizes D. Hint: describe D more simply. Justify that the simpler description defines the same language.

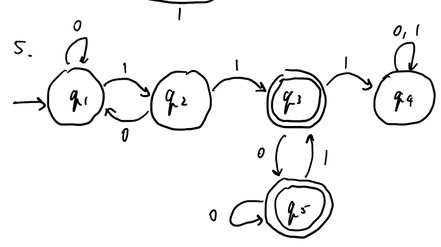
① 0. 

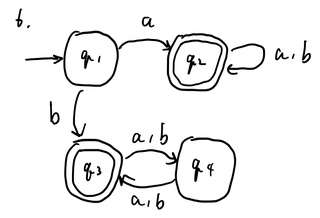
1. 

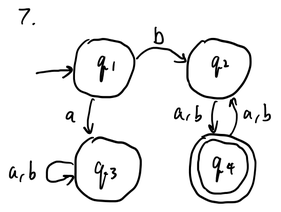
2. 

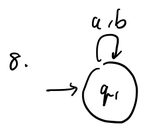
3. 

4. 

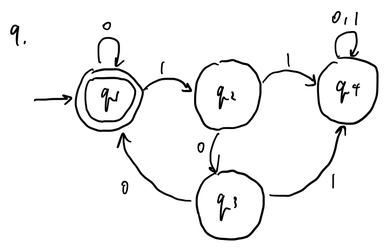
5. 

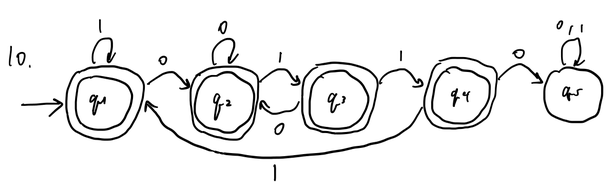
6. 

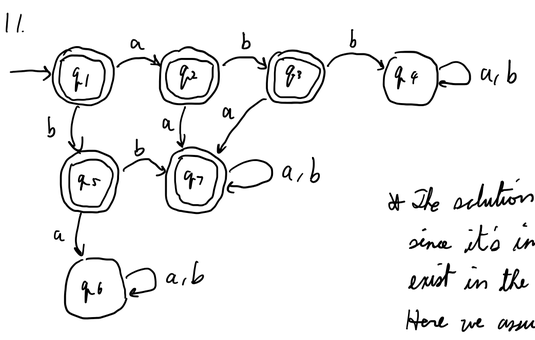
7. 

8. 

$L_1 \cap L_2 = \phi$

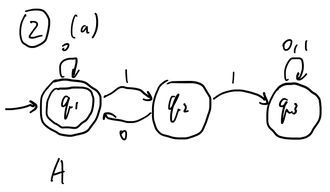
9. 

10. 

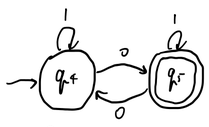
11. 

→ 0

* The solution can also be $L = \{w \in \{a,b\}^*\}$ since it's impossible to have abb **and** ba exist in the same time. Here we assume the question meant abb **or** ba

② (a) 

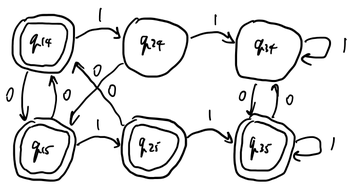
A

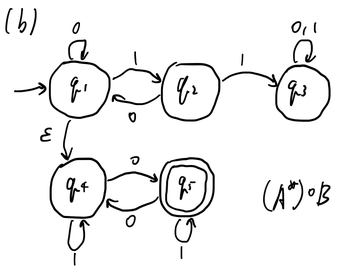


B

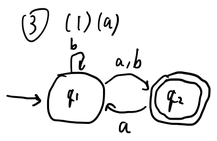
A ∪ B

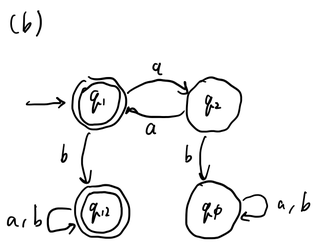
	Σ	0	1
	q#		
accept/start	14	15	24
accept	15	14	25
	24	15	34
accept	25	14	35
	34	35	34
accept	35	34	35

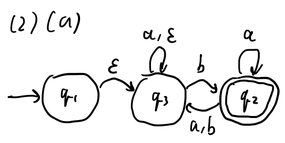


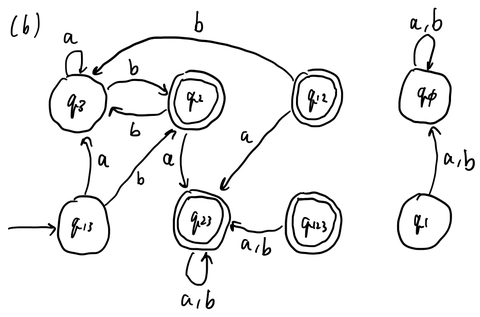
(b) 

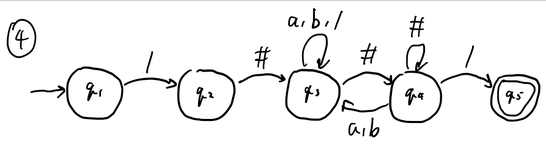
$(A^*) \circ B$

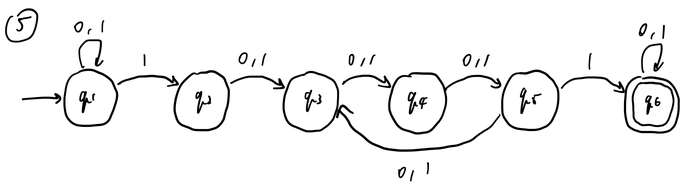
③ (1) (a) 

(b) 

(2) (a) 

(b) 

④ 

⑤ 

⑥ Thought Process

- 5 states means composition of DFA won't work ($2 \cdot 2 \cdot 3 = 12 > 5$)
- It must be represented by ≤3 NFA states ($2^2 < 5 < 2^3$, $4 < 5 < 8$)
- Not containing ab implies w can only be "a...." or "b...a..." or "b..."

D = a... or b...a... or b...

= $b^* \circ a^*$ odd implies b cannot be 0.

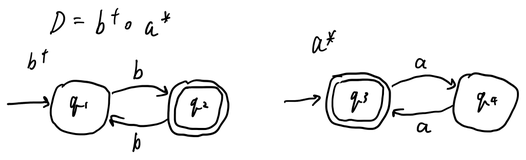
odd even

= $b^+ \circ a^*$ this eliminates a... as b must exist in w.

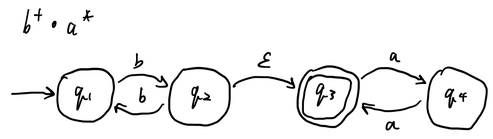
odd even

Solution

$D = b^+ \circ a^*$



$b^+ \circ a^*$



To DFA with $Q' = \{q_1, q_{23}, q_4, q_\phi\}$, since only q_{23} & q_ϕ are involved

